DeepFake Analysis Report

Explainable Deepfake Detection — Xception-LSTM with LIME · Research prototype

Analysis ID	39b0107a-3b83-4d09-93ae-a6b48391e20a
Generated	2026-11-04 16:18 UTC
Analyzed at	—
Status	—

1. Executive summary

	raw fake score	decision threshold	model version
—	0.465165	0.75 (hf-space-original)	—

This score is an uncalibrated sigmoid output — it is not a probability and not forensic proof.

2. Media information

Filename	test_video.mp4
Media type	—
File size	164.2 KB
Resolution	1920×1080
Duration	0.25 s
FPS	24.0
SHA-256	b3d47c8919ae58ae2e0ca219b04a59f1dadfe72d37519087c03dd9d371124a49
Frames extracted / analyzed	5 / 5

3. Prediction analysis

Raw model output (fake score)	0.465165
Decision method	Sliding-window ensemble (full 32-frame score + 10-frame windows, averaged)
Decision threshold	0.75 (hf-space-original)
Score calibrated?	No — sigmoid output, uncalibrated
Model version	—
Window score variation	— (no window scores)

5. Visual evidence



Processed face (elliptical mask applied)

LIME was not requested or is unavailable for this analysis.

LIME highlights regions of a single replicated frame that influenced the model. It does not prove manipulation and does not explain the temporal pathway.

6. Technical details

Model	Xception-LSTM · version —
Checkpoint SHA-256	—
Sequence length	— frames @ —×— px
Preprocessing	OpenCV DNN face detection (confidence 0.5) → 20% margin crop → 299×299 resize → elliptical background mask → [0, 1] normalization
Prediction method	Sliding-window ensemble (full 32-frame score + 10-frame windows, averaged)
Processing duration	—

7. Interpretation

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9. Research limitations

1. Scores are sigmoid outputs from an uncalibrated model; they are not proven probabilities.

2. Models were trained only on the DeepFake Detection (DFD) subset of FaceForensics++; performance on other manipulation methods is unknown.

3. Only the most confident face per frame is analyzed; multi-face videos are not fully covered.

4. LIME explains a replicated static frame; it does not attribute the temporal (LSTM) pathway.

5. Face-detection variability across OpenCV builds can change scores for the same media.